

Yusuf Hasan

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Electrical Engineering

Vancouver, BC

Expected: November 2026

EXPERIENCE

Software Development Engineer Intern

Jun 2026 – Aug 2026

Amazon

Vancouver, BC

- Sole engineer on a **full-stack self-service configuration platform** adopted by **40+ clients**, surfacing profiles — collections of keys and plugin/selector functions invoked on actions — stored across hierarchical datatables with fallback and inheritance
- Defined API in **Smithy IDL**; implemented read service in **Java on AWS Lambda** with cascading lookup across **7 filter dimensions** — **~64 ms** reads against $p50 \leq 500 \text{ ms}$ / $p95 \leq 1.5 \text{ s}$ SLOs
- Built **React + TypeScript** SPA with **cascading dropdowns** and a **pluggable API-client layer**, decoupling frontend delivery from backend readiness
- Provisioned cloud infra as **AWS CDK** (Lambda, VPC, Route 53, IAM); **99+ unit/integration tests** with schema **drift-guard coverage**, shipped via peer-reviewed CI/CD

Hardware Pre-Development Intern

June 2024 – April 2025

Robert Bosch GmbH (Bosch)

Reutlingen, Germany

- Designed **STM32**-based test/data-collection **PCBs** in **Altium** — onboard memory, multi-protocol interfaces (FDCAN, SPI, I²C, UART, USB3300), **12V→5V buck + 3.3V LDO** — enabling evaluation of **next-gen MEMS inertial sensors** at $>12.8 \text{ kHz}$
- Wrote **C firmware** for concurrent data collection on 10+ **automotive MEMS sensors** during field testing
- Characterized next-gen IMU performance (bias, noise, offset, sensitivity) across vibration profiles using **MATLAB** and **Python**
- Developed a modified **rate-table** maytagging methodology in **Python** to characterize MEMS gyroscope **offset g-sensitivity**, accounting for Earth rotation

PROJECTS

Hybrid Localization Unit | C, Python, ESKF, PCB Design, IMU, UWB

Sep 2025 – Present

- Designed **HYLO** — a custom **STM32 PCB** integrating a DWM3000 UWB module and MEMS IMUs — for real-time indoor localization in GNSS-denied environments
- Modelled and tuned an **Error-State Kalman Filter (ESKF)** in **C**, fusing high-rate IMU prediction with asynchronous UWB range corrections on embedded hardware

2-DoF Laser Projector System | Altium, Embedded C, SolidWorks, FPGA, Verilog

Feb 2024 – April 2024

- Built a motor-controlled laser projector in a **team of 4**; programmed a quadrature-encoder decoder in **SystemVerilog** for closed-loop position feedback
- Programmed an **ESP32 ISR** in **C** for **SPI** communication and dual motor control; implemented a real-time **PID controller** projecting multiple animations

RC4 Decryption and Cryptanalysis Accelerator | SystemVerilog, FPGA, FSM Design

June 2025

- Designed a hardware RC4 stream cipher accelerator in **SystemVerilog**, implementing the **KSA** and **PRGA** across coordinated finite-state machines
- Automated brute-force sweep of **16.7 M 24-bit keys** with real-time plaintext validation; optimized for FPGA with **dual-port memory** and **pipelined FSMs**

TECHNICAL SKILLS

Software: C, Java, Python, TypeScript, MATLAB, SystemVerilog, 8051 Assembly, ARM Assembly, React

Cloud & Tools: AWS (Lambda, CDK, VPC, IAM), Git, CI/CD, STM32CubeIDE, Quartus, Simulink, TensorFlow, PyTorch

Electrical: Altium, Cadence Virtuoso, KiCad, LTspice, PSIM, SolidWorks, Logic Analyzer, Function Generator, Oscilloscope